Vietnam National University Ho Chi Minh City
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Deep Learning on Large-Scale Data and Specialized Tasks

CO3133 - Deep Learning and Its Applications
Assignment 2
Group: Placeholder

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Content

1. Data Description
2. Methodology
3. Implementation Results

SECTION 01

Data Description

SUIM Dataset

Objective: Recognize what surrounds a robot underwater, pixel by pixel.

Data: 1,635 image-mask pairs (1,525 train-val, 110 official test). After the audit, 1,540 remain: 1,143 train, 287 validation, 110 test.

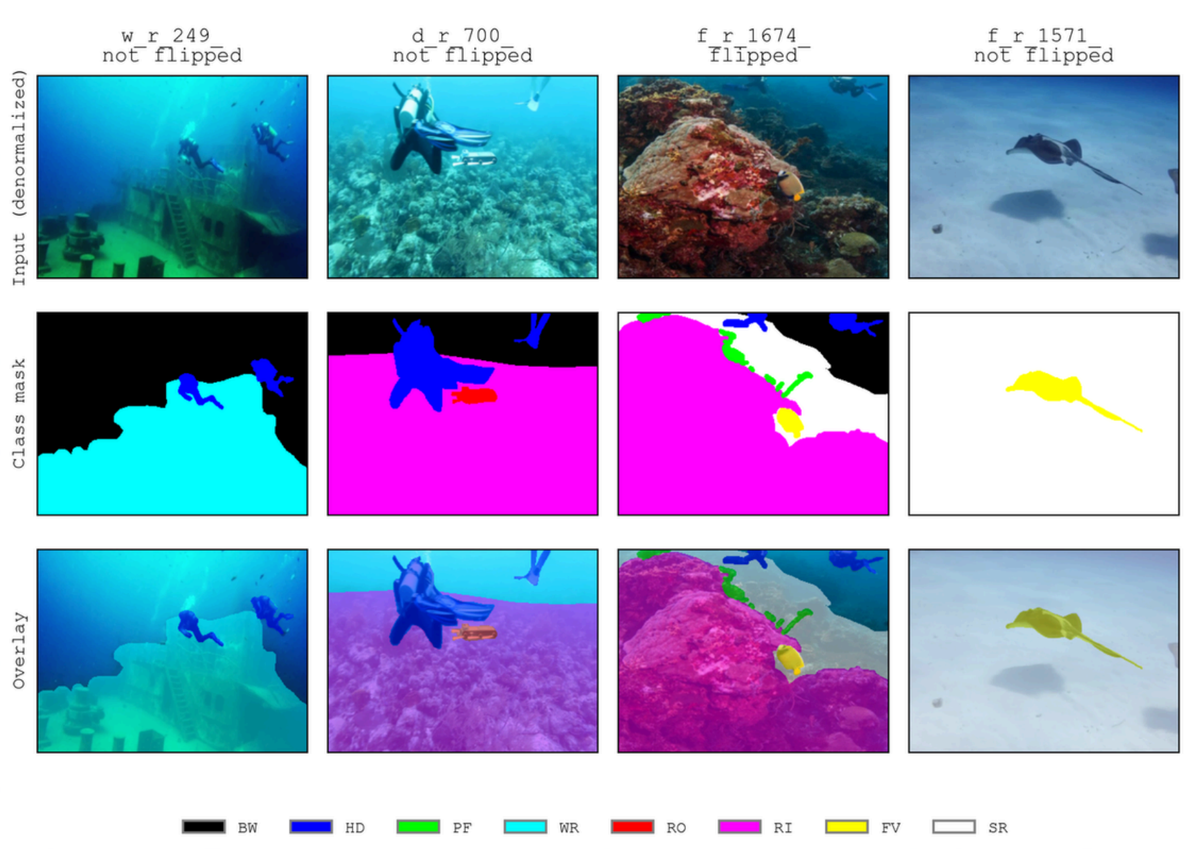


Figure 1. Training samples with their masks and overlays.

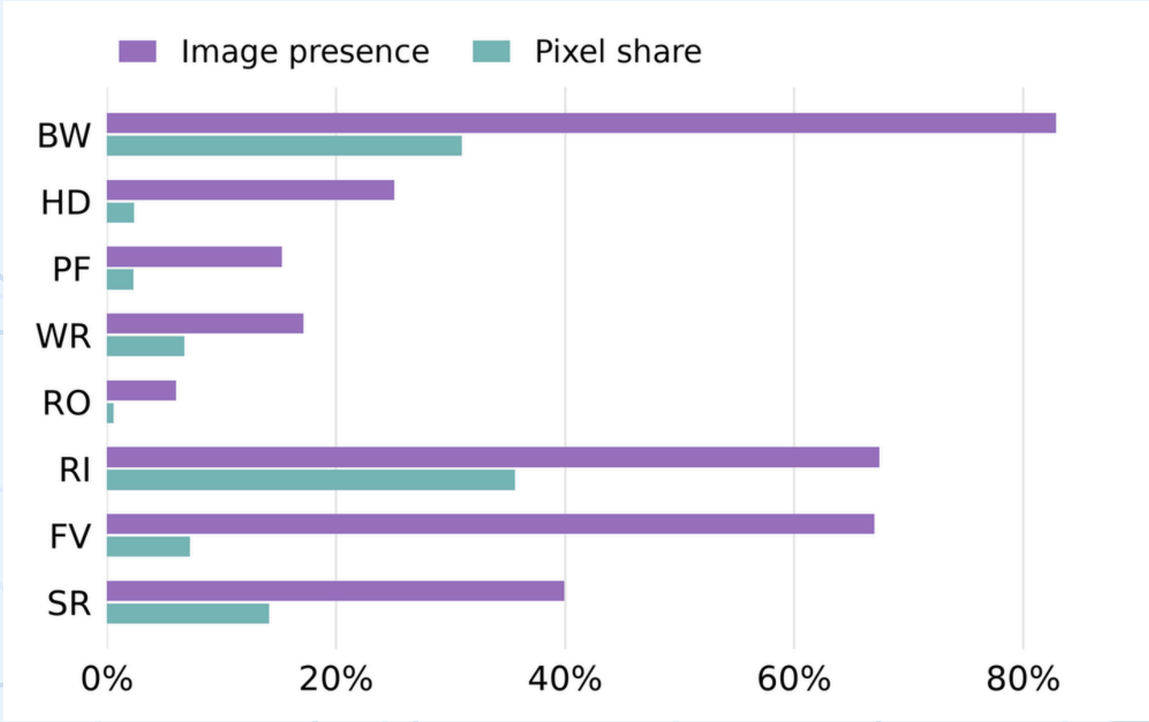


Figure 2. Class imbalance: share of images and of pixels per class.

SUIM dataset: EDA

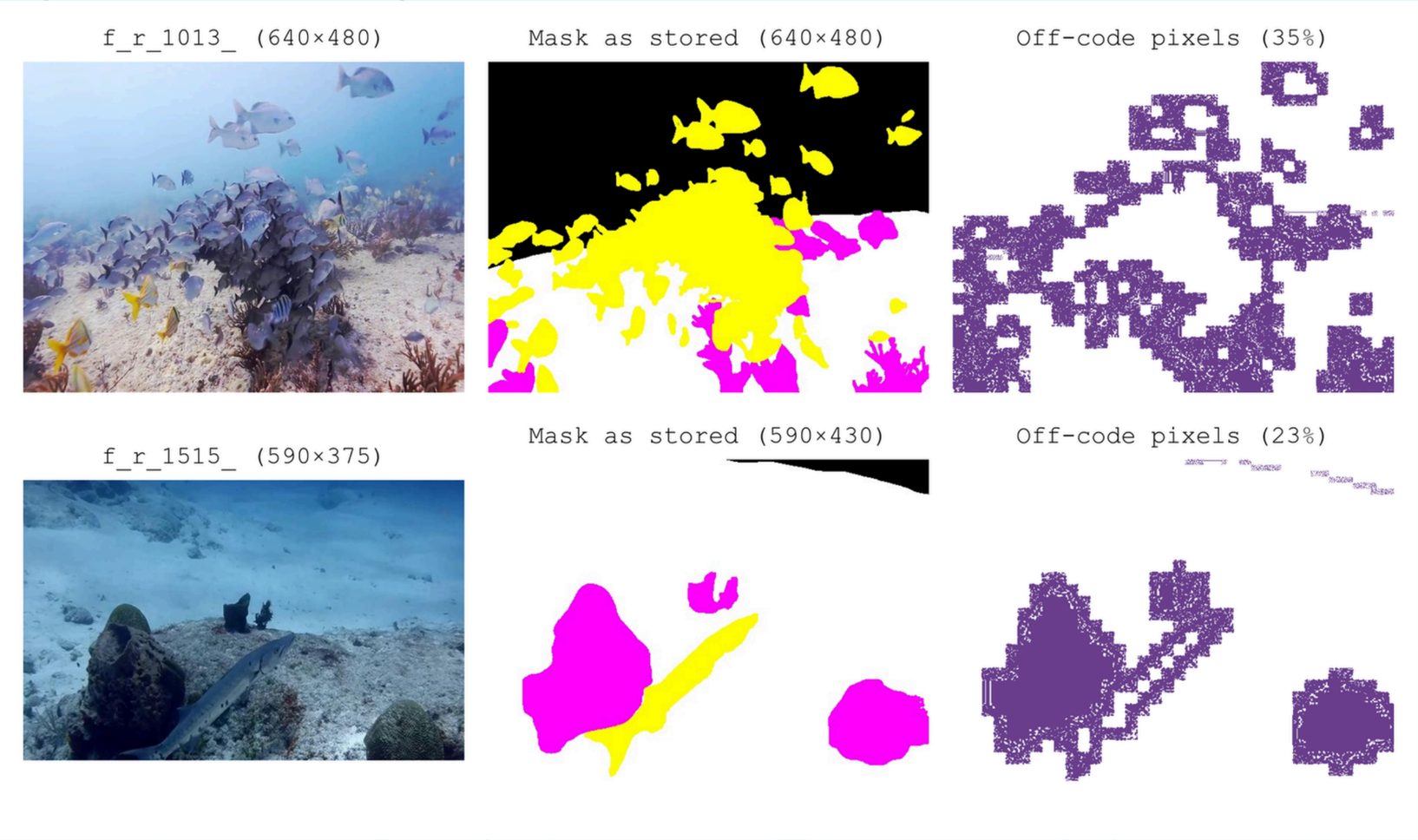


Figure 3. Masks with off-code (invalid) colors. 85 masks in the training pool contain them and are excluded.

SUIM dataset: EDA

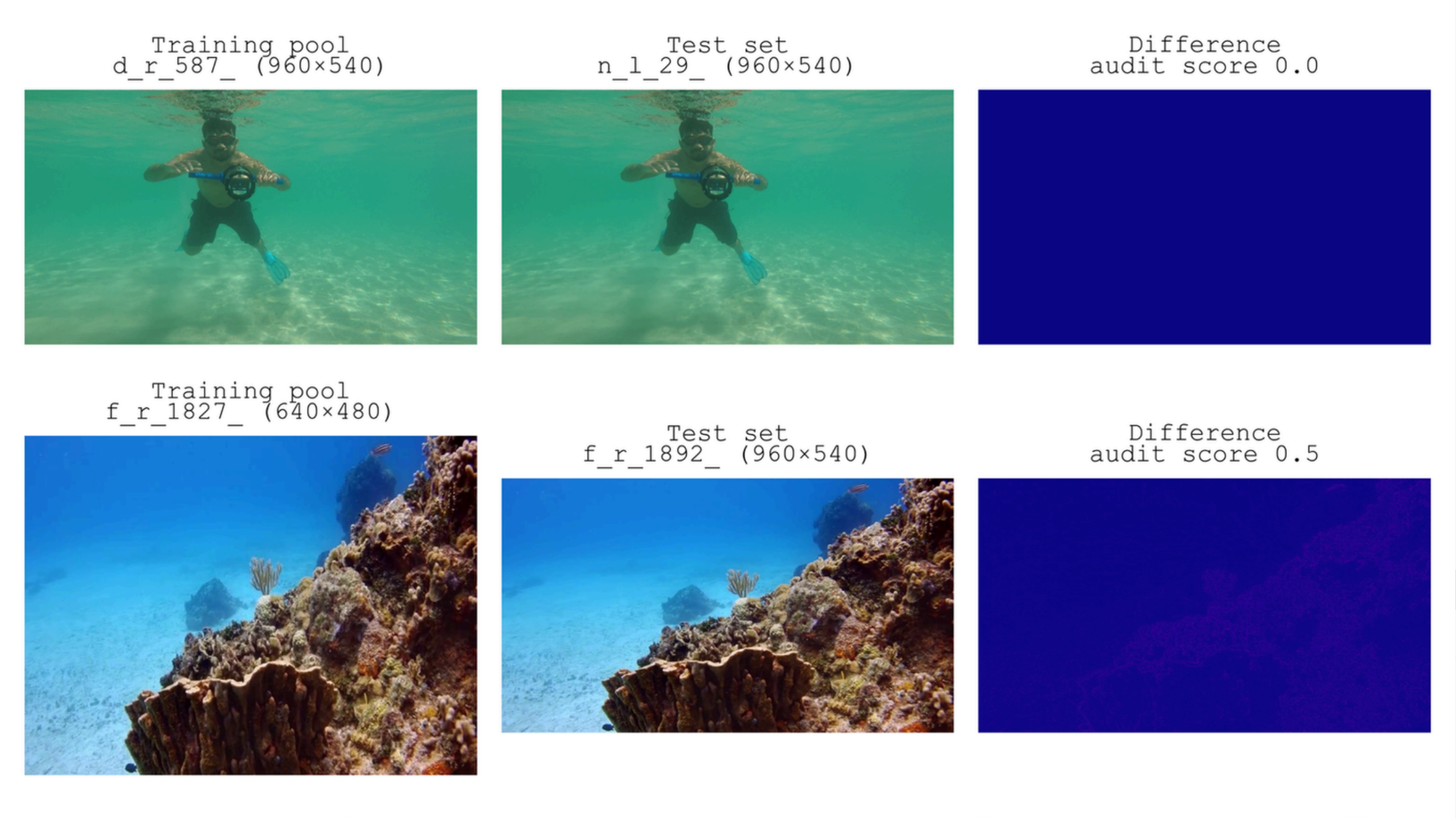


Figure 4. Duplicate photos between the training pool and the test set, scored by pixel difference. The 10 training copies are removed.

SUIM dataset: EDA

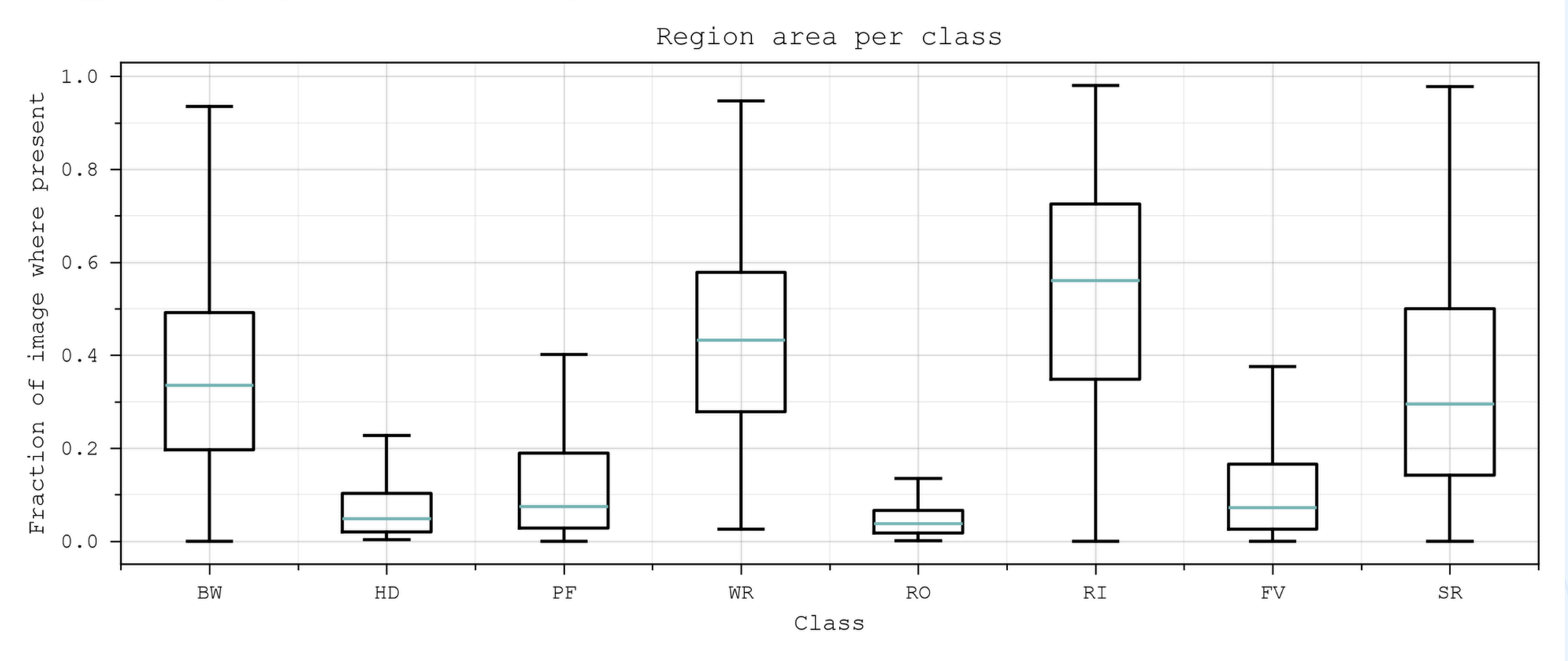


Figure 5. Share of the image covered by each class when it is present. Divers, plants, robots and fish usually cover small regions.

SUIM dataset: EDA

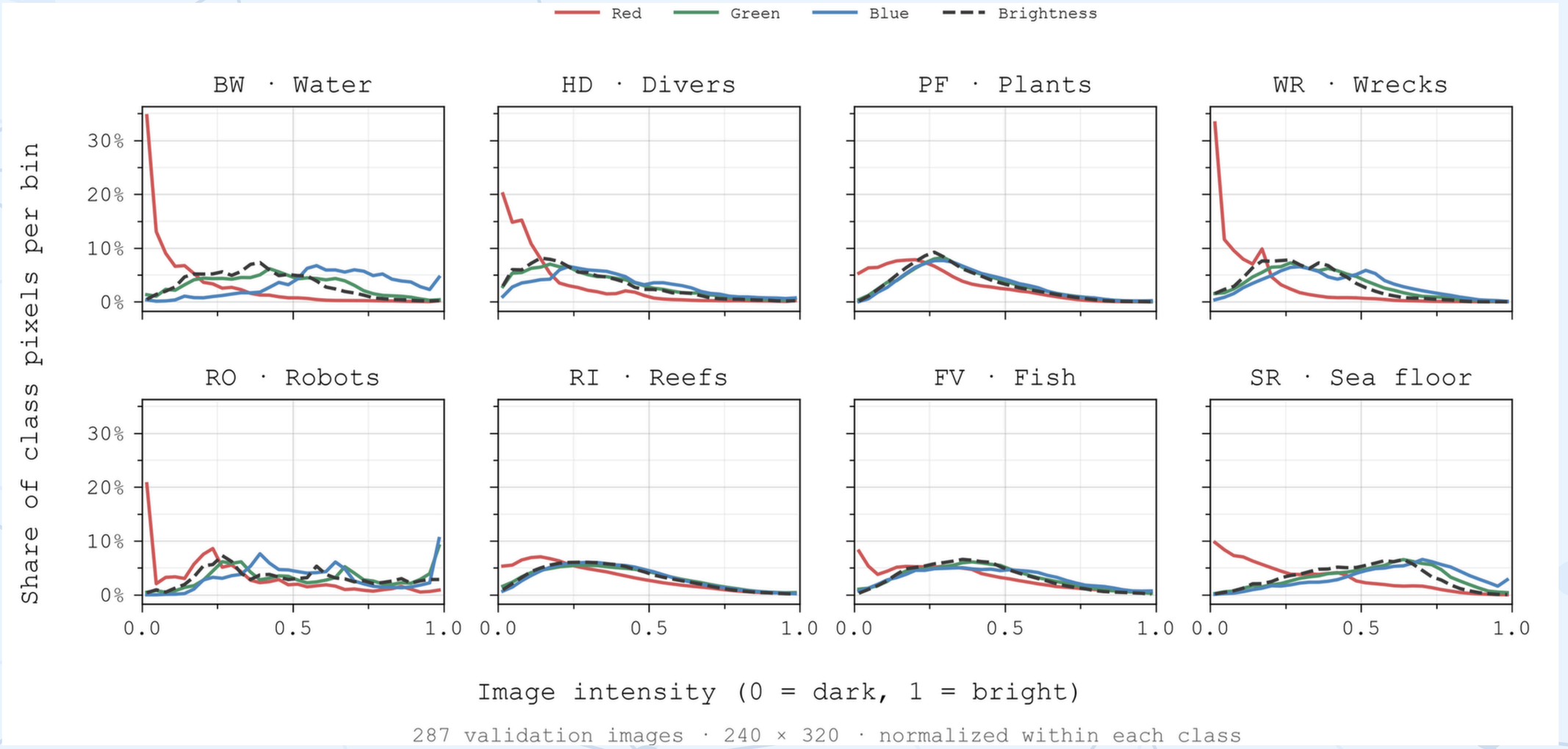


Figure 6. Color (RGB) and brightness distributions per class. Plants (PF) look similar to reefs (RI) and fish (FV), and robots (RO) have no consistent color.

SUIM dataset: EDA

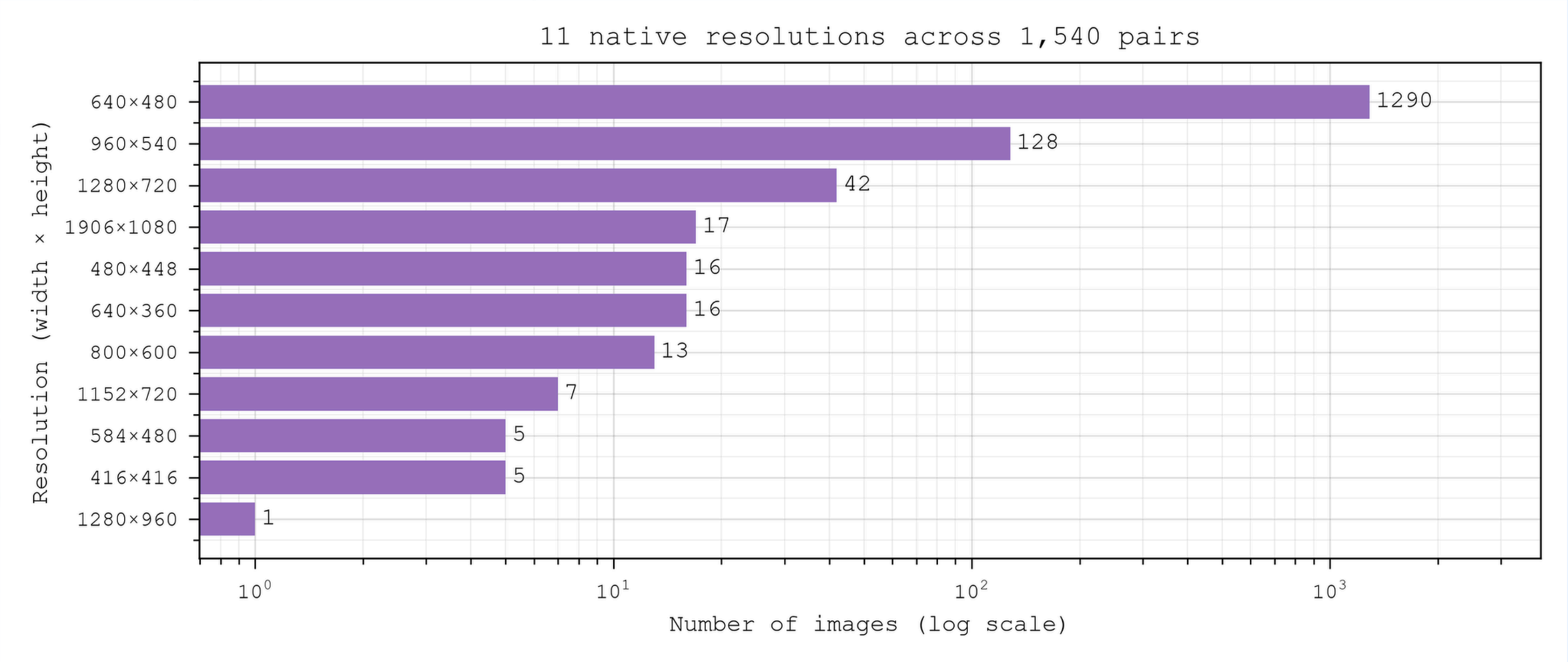


Figure 7. Native resolutions. Most images are 640 x 480, an aspect ratio of 4:3.

SUIM dataset: Preprocessing

Preprocessing:

1. Decode the RGB masks into class labels 0 to 7
2. Resize to 240 x 320: bilinear for images, nearest neighbour for masks
3. Flip image and mask together (p = 0.5), training only
4. Normalize pixels with the ImageNet mean and std
5. Group the pairs into batches

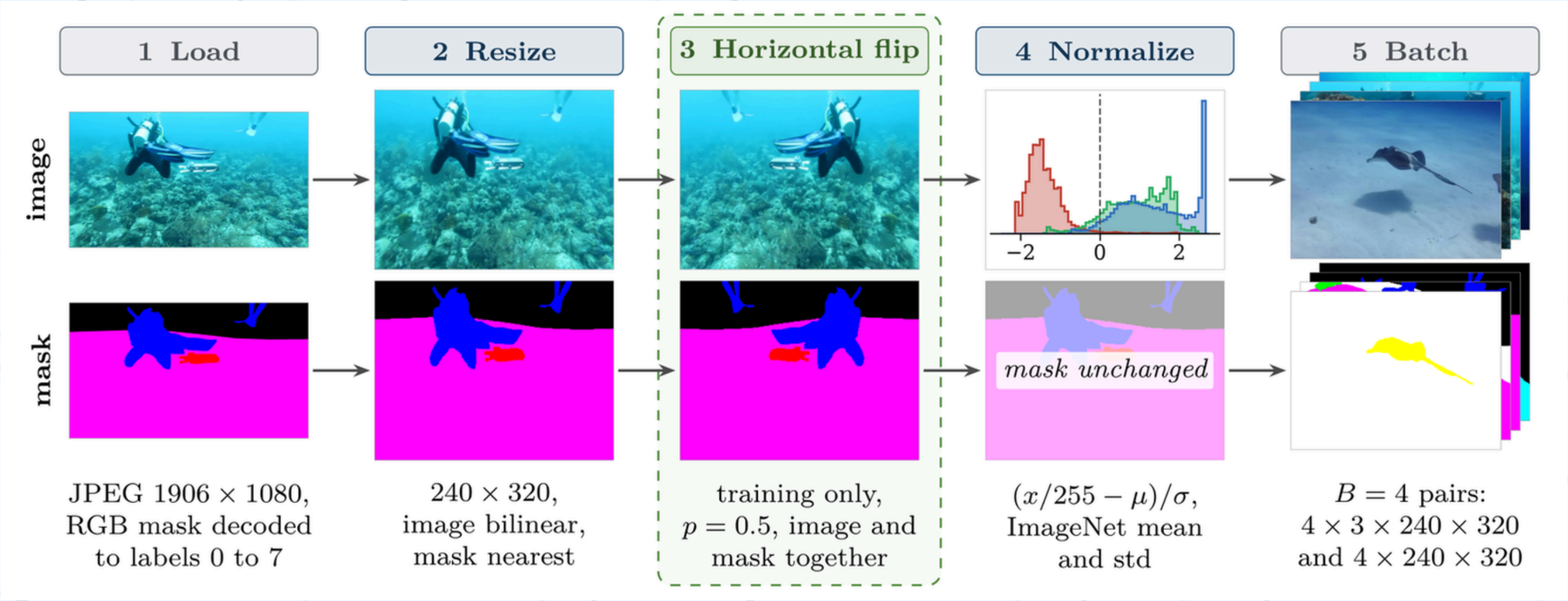


Figure 8. Preprocessing of one training pair, from the raw files to a batch.

SECTION 02

Methodology

Method: UNet

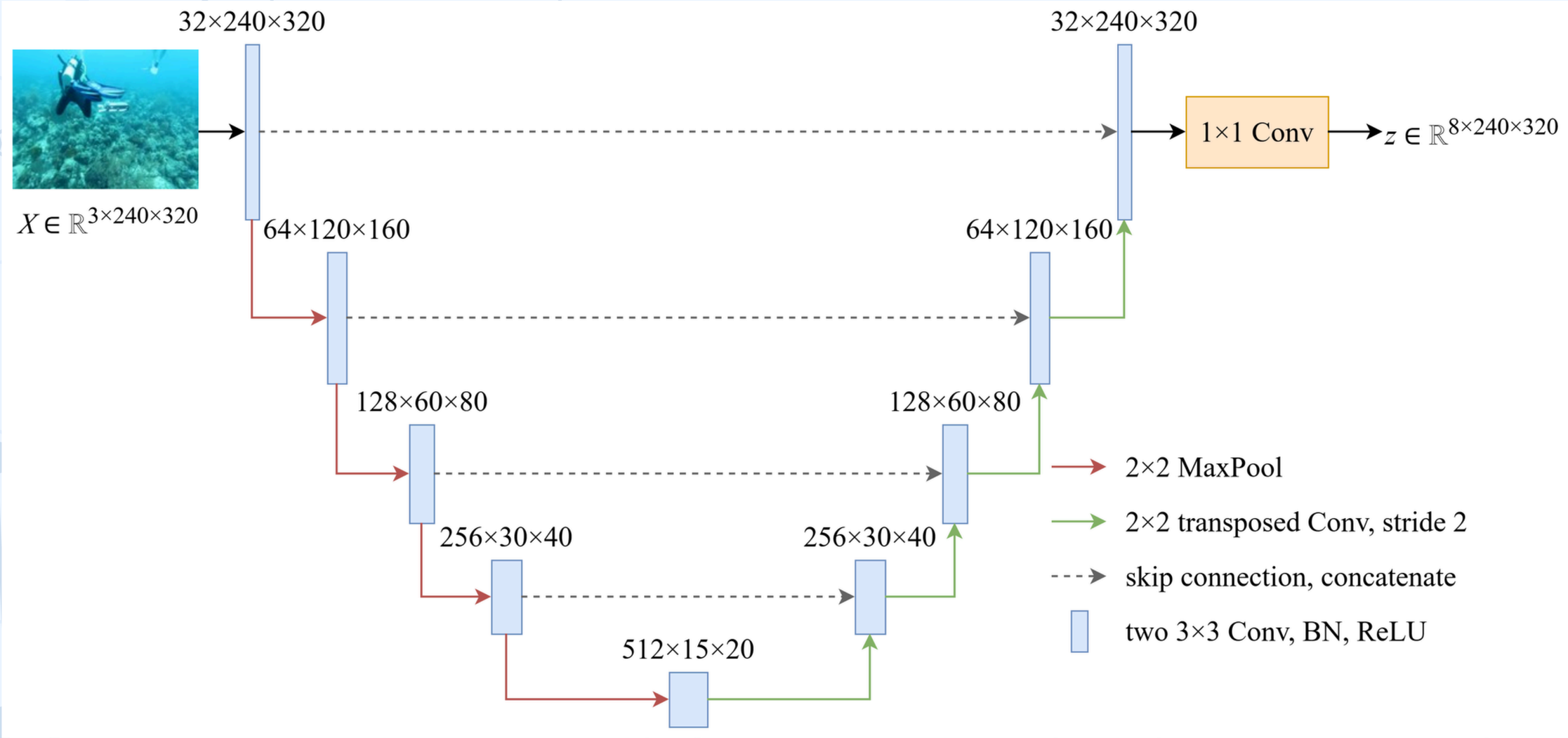


Figure 9. UNet architecture as our baseline.

Method: DeepLabV3

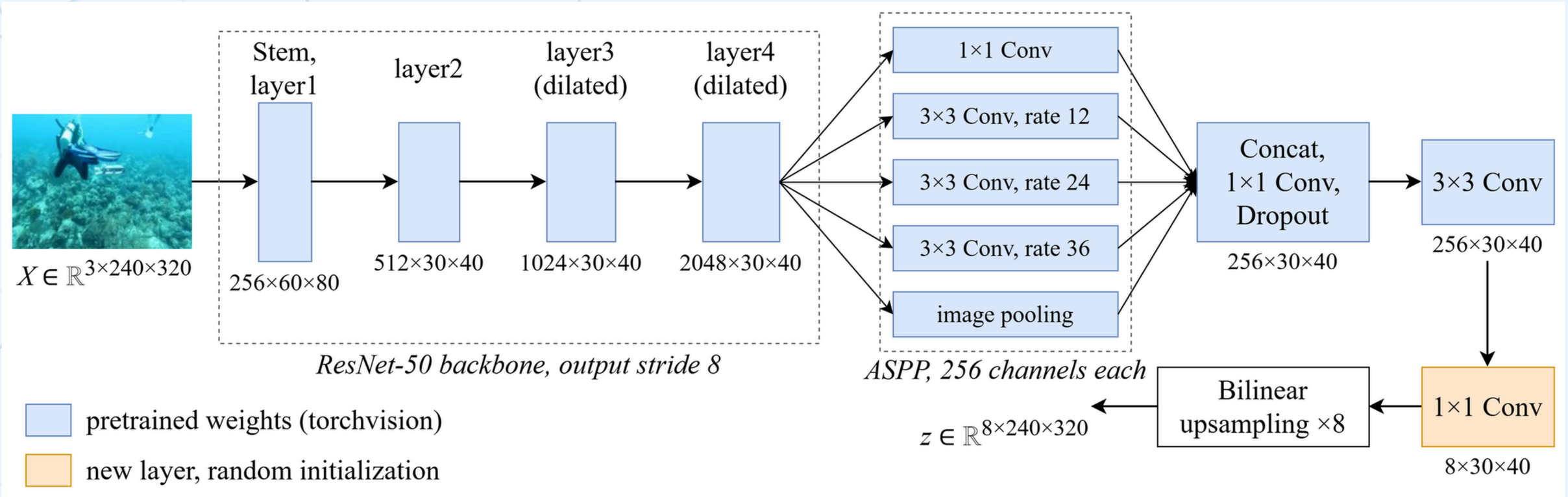


Figure 10. DeepLabV3 architecture as our pretrained backbone.

Loss and evaluation

Training

Soft Dice:
Averaged over classes to learn regions

$$D_{n,k} = \frac{2\sum_i p_{i,k}\, g_{i,k} + \epsilon}{\sum_i p_{i,k} + \sum_i g_{i,k} + \epsilon},$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{1}{BK}\sum_{n=1}^{B}\sum_{k=1}^{K} D_{n,k},$$

CE loss:
Averaged over pixels, to learn the correct class

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{N} \log p_{i,y_i},$$

Evaluation

mIoU:
Measure intersection over union

$$\text{mIoU} = \frac{1}{K}\sum_{k=1}^{K} \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k + \text{FN}_k}$$

Dice Score:
Like mIoU, but counts the overlap twice

$$\text{Mean Dice} = \frac{1}{K}\sum_{k=1}^{K} \frac{2\text{TP}_k}{2\text{TP}_k + \text{FP}_k + \text{FN}_k}$$

Note: *All models train with CE; the loss experiment uses 0.5 CE + 0.5 soft Dice. Each run keeps the checkpoint with the highest validation mIoU.*

SECTION 03

Implementation Results

Training Results

UNet

DeepLabV3

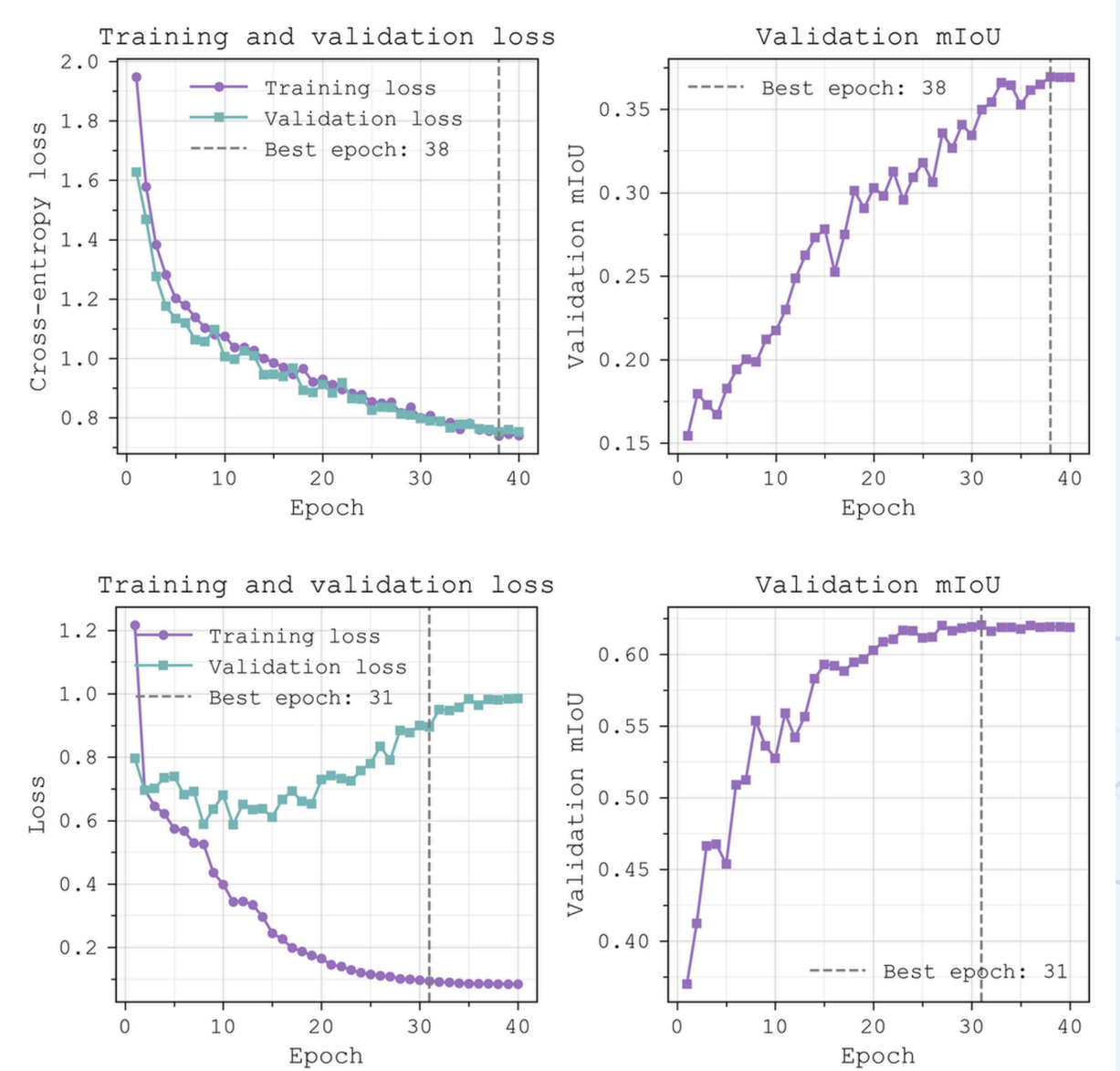


Figure 11. Training curves of U-Net (top) and DeepLabV3 with CE (bottom).

Quantitative Results

		Validation (287 images)		Test (110 images)		
Model	Loss	mIoU (%)	Mean Dice (%)	mIoU (%)	Mean Dice (%)	Params (M)
U-Net	CE	36.94	48.05	36.75	47.84	7.76
DeepLabV3	CE	62.05	73.91	68.27	79.47	39.64
DeepLabV3	CE + Dice	62.28	74.42	69.18	80.78	39.64

Table 1. Validation and test results at the checkpoints selected on validation. Seed 42; each model was scored once on test.

Findings:

- DeepLabV3 (CE) beats U-Net by 31.52 points of test mIoU (25.11 on validation), with 5.1 times the parameters.
- DeepLabV3 scores higher on test than on validation: the test split has a different class mix (fewer reef images). Test images are not closer to the training set than validation images.
- CE + Dice changes mIoU by only +0.23 (validation) and +0.91 (test), less than the 0.97-point gap between two identical training runs.

Class-Level Results

Model	Split	BW	HD	PF	WR	RO	RI	FV	SR
U-Net (CE)	Validation	80.62	30.83	0.00	41.16	0.00	67.94	33.14	41.85
	Test	84.82	33.75	0.05	33.30	0.00	54.31	31.64	56.12
DeepLabV3 (CE)	Validation	83.89	70.32	11.94	66.88	70.58	75.36	66.89	50.56
	Test	89.43	78.96	24.12	70.32	71.09	69.57	70.08	72.58
DeepLabV3 (CE + Dice)	Validation	83.84	70.17	14.87	66.42	68.20	75.28	67.13	52.30
	Test	90.04	77.09	34.78	69.53	68.87	69.40	70.44	73.30

Table 2. Per-class IoU (%) on the validation and test splits at the selected checkpoints.

Findings:

- DeepLabV3 beats U-Net on every class on both splits. U-Net gets almost no plant (PF) or robot (RO) pixels right.
- Plants stay the hardest class: 24.12% test IoU for DeepLabV3 (CE).
- CE + Dice on test: PF +10.66, RO -2.22, WR -0.79 points. Validation shows the same pattern, so our hypothesis that Dice improves all three rare classes is not supported.

Qualitative Results

Robot Example

Best Example

Worst Example

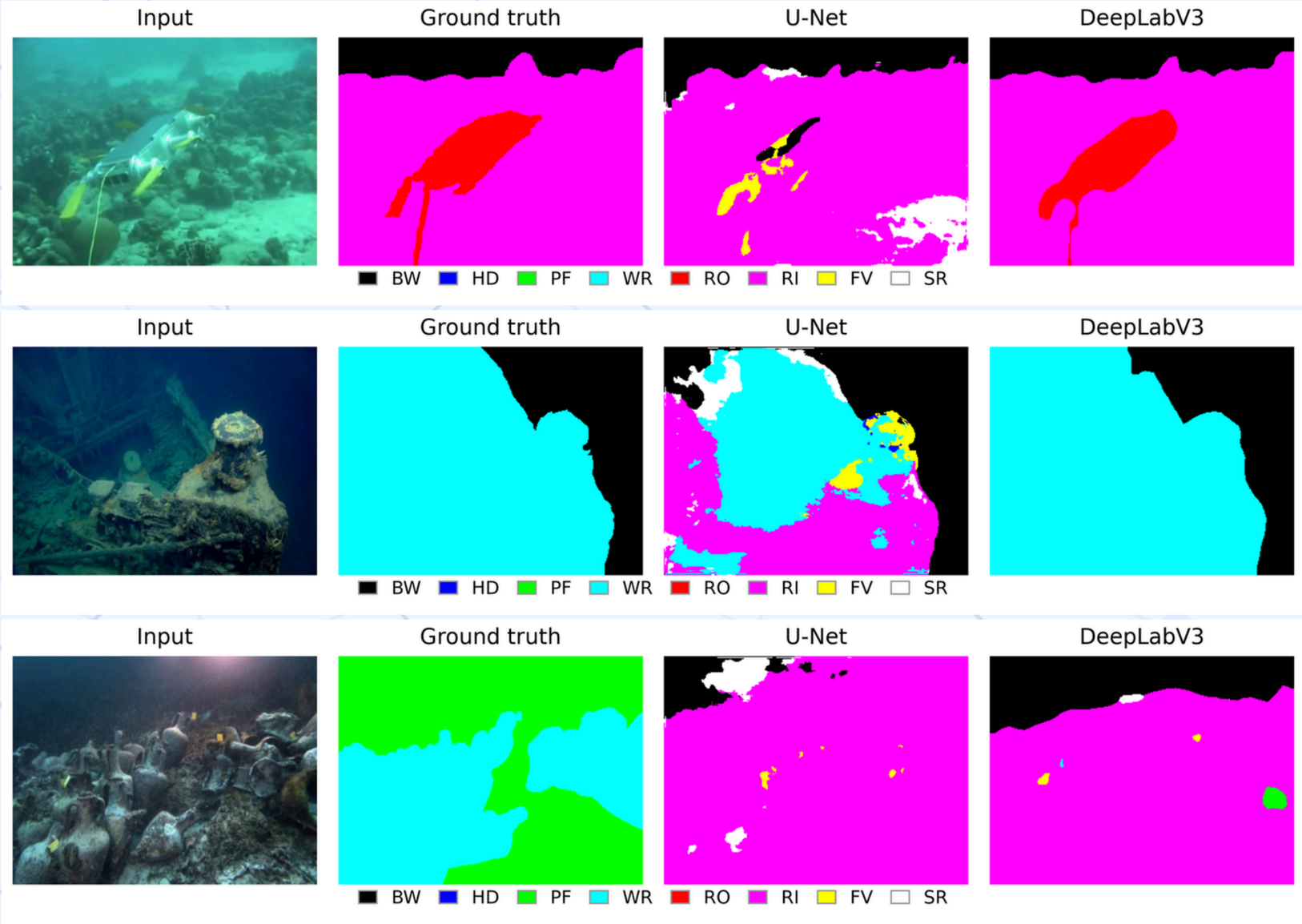


Figure 12. U-Net and DeepLabV3 (CE) on the same validation images.

Compute Cost Analysis

Table 3. Training cost and inference time of U-Net and DeepLabV3.

Model	Params (M)	GFLOPs	Train time (min)	Latency (ms/image)	Peak memory (GiB)	Training hardware
U-Net	7.76	28.3	14.9	4.8	0.97	RTX 4060 Laptop
DeepLabV3 (CE)	39.64	95.8	21.4	7.8	1.93	RTX 4060 Ti
DeepLabV3 (CE + Dice)	39.64	95.8	21.6	7.8	1.93	RTX 4060 Ti

Trade-off:

- U-Net is the smallest and fastest model (4.8 ms/image), but far less accurate.
- DeepLabV3 has 5.1 times the parameters and 3.4 times the FLOPs and needs 7.8 ms/image. That still fits a 10 ms budget if other steps (e.g. streaming) add little latency.
- Adding Dice changes neither the parameters nor the inference cost.

Controlled Results

Settings:

- **Reference:** DeepLabV3 (CE), fine-tune ResNet50 and head
- Frozen backbone, only fine-tune the head
- No augmentation
- ResNet101 backbone
- CE + Dice loss
- Half resolution: 120 x 160

Findings:

- Clear effects: frozen backbone -7.65 and half resolution -5.76 mIoU points.
- The other changes are at most 0.35 points, no more than retraining the same reference.

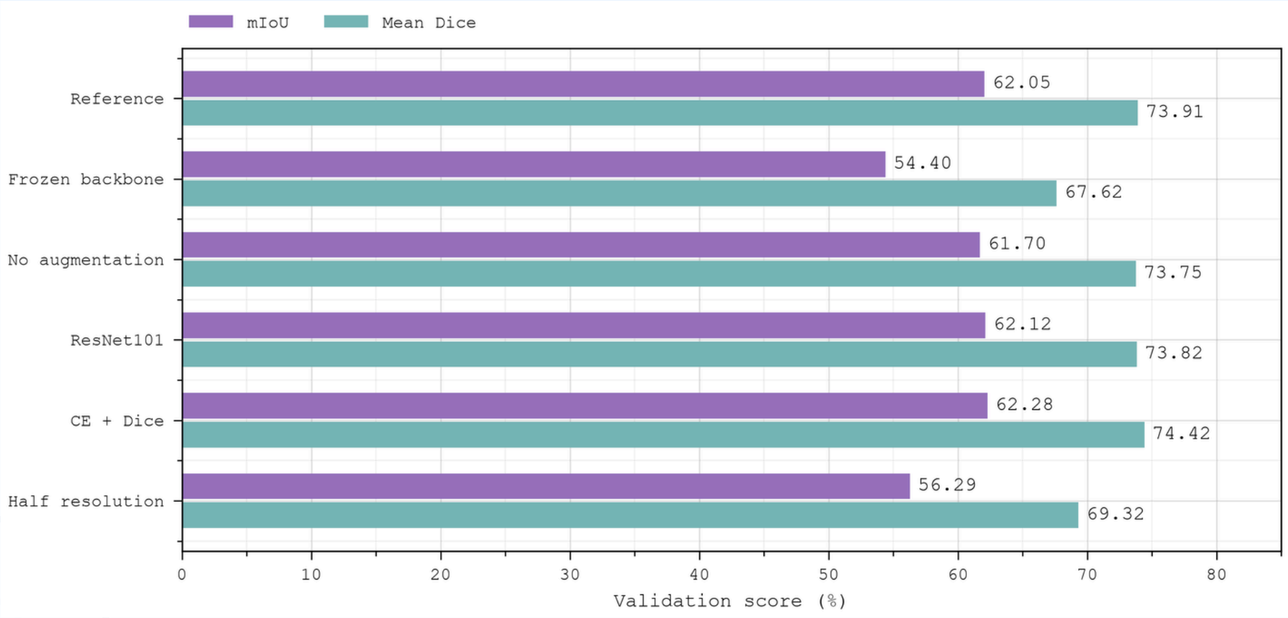


Figure 13. Validation mIoU and mean Dice of the controlled runs.

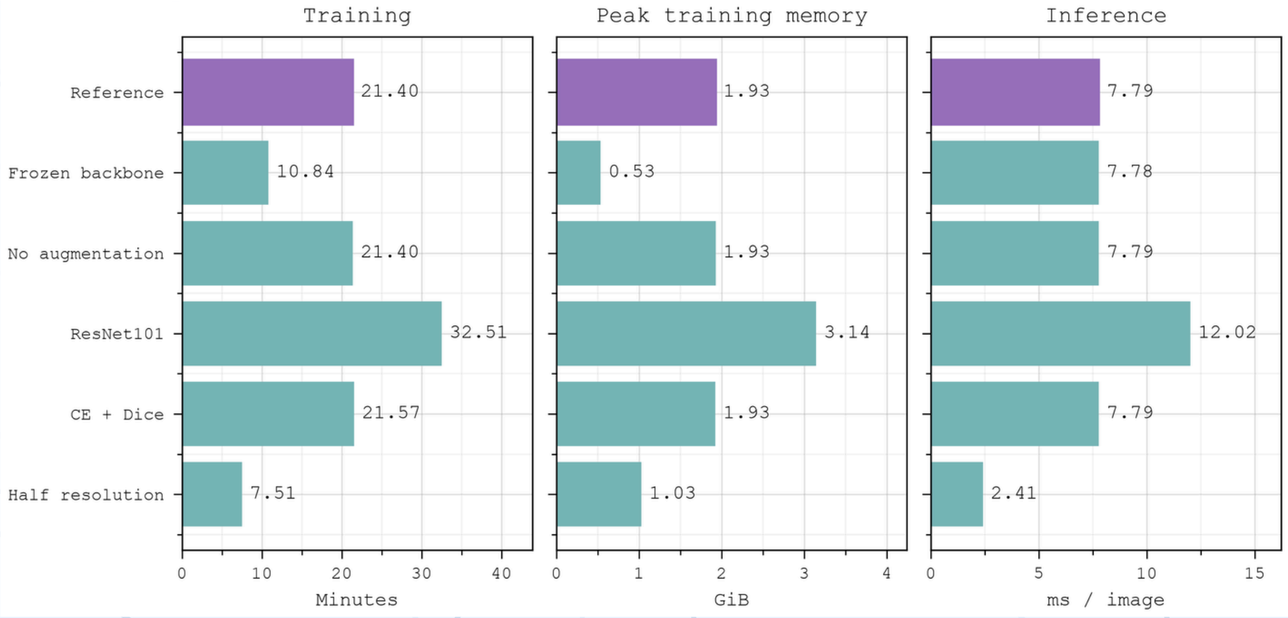


Figure 14. Training time, peak memory and inference latency.

Error Analysis

Most confused: 71.26% of plant pixels go to reef, 11.46% of robot pixels to divers.

Boundaries are hard: pixels on either side of a class edge reach 30.07% mIoU for DeepLabV3, against 64.34% inside regions (U-Net: 18.04% vs 37.88%).

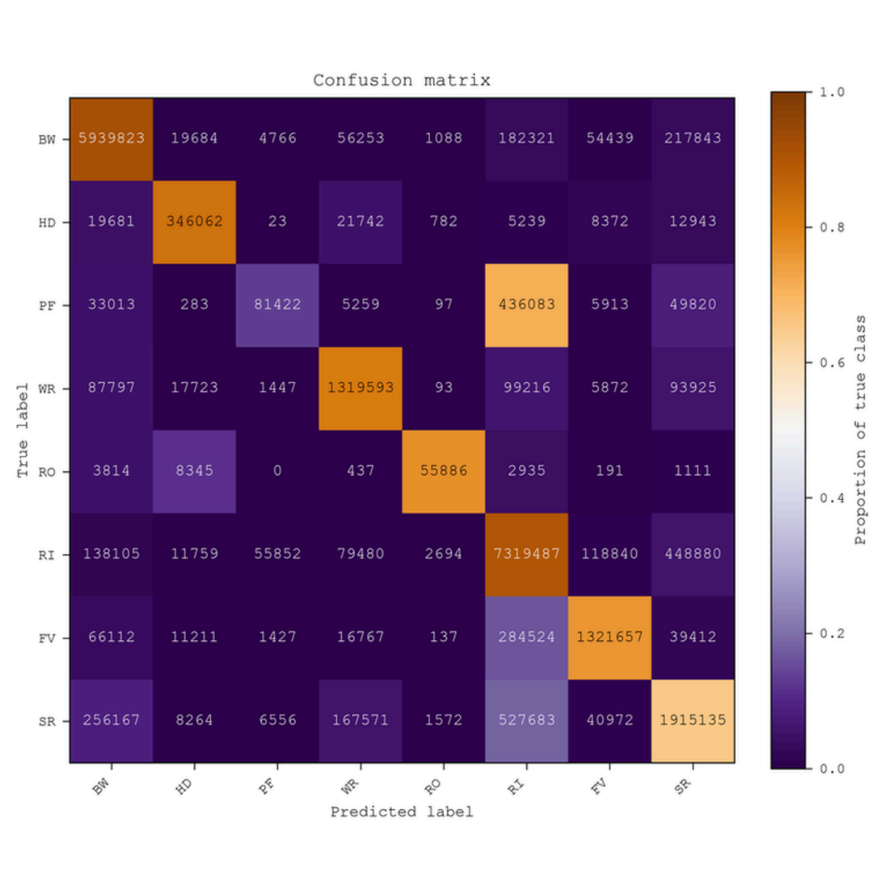


Figure 15. Validation confusion matrix of DeepLabV3 (CE), normalized per true class.

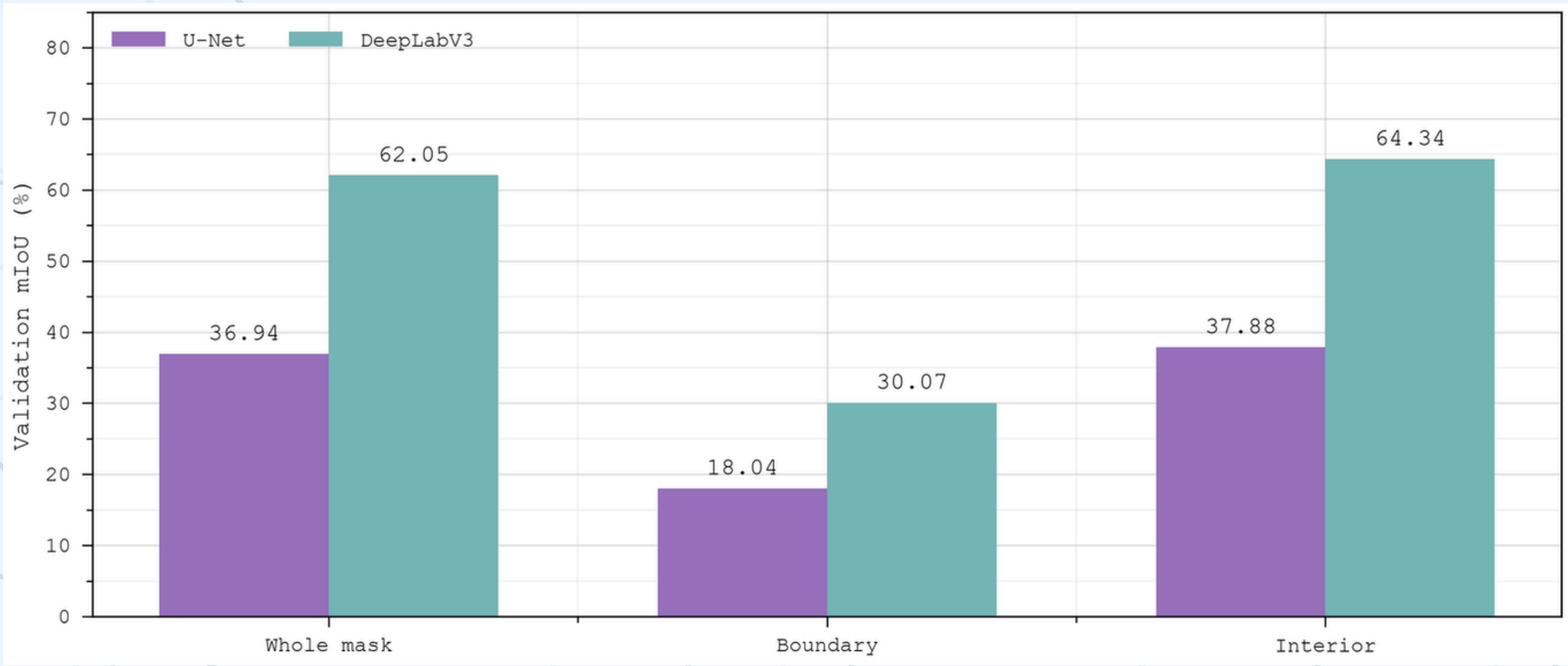


Figure 16. Whole-mask, boundary and interior validation mIoU.

Error Analysis

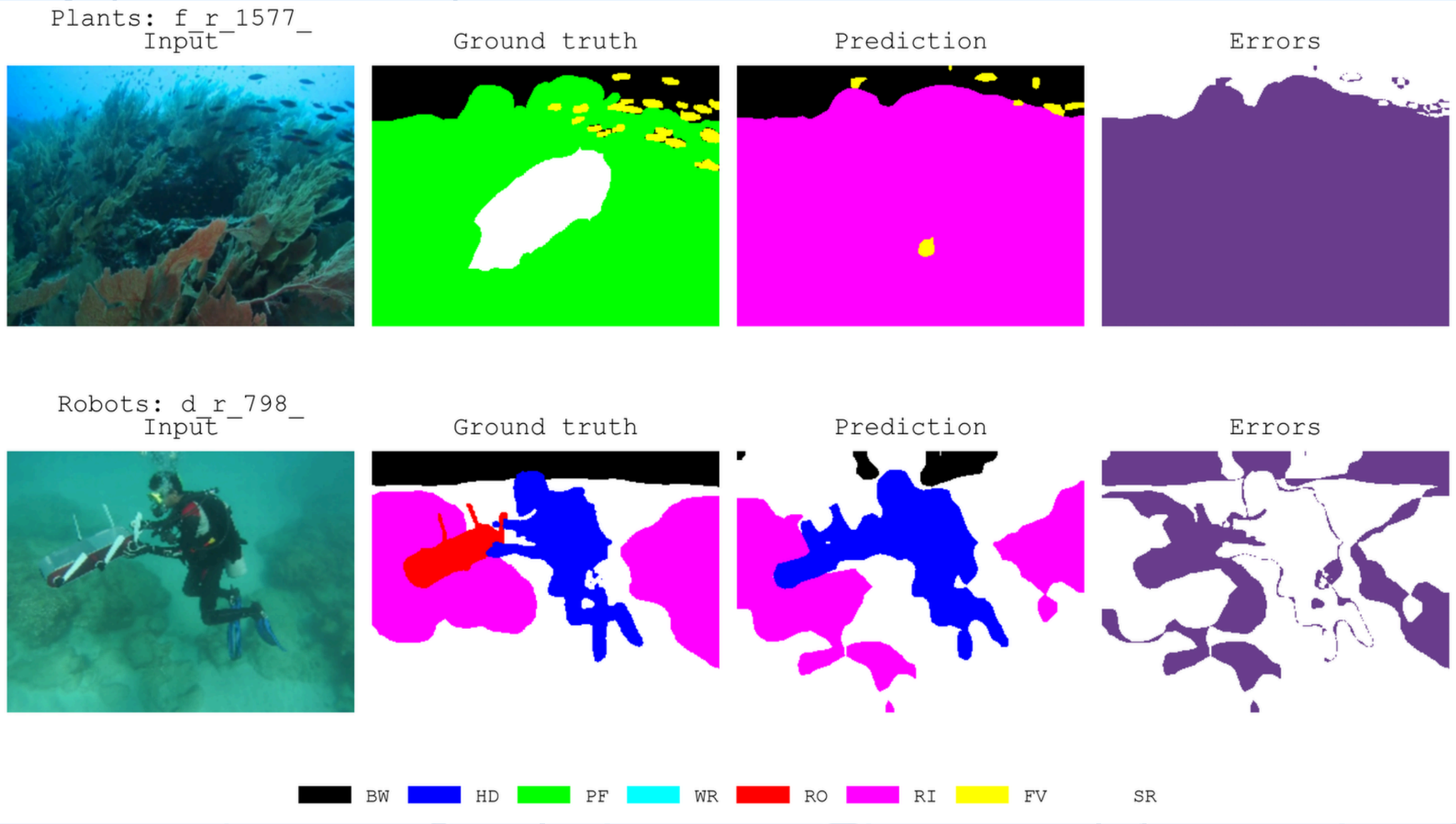


Figure 17. DeepLabV3 (CE) failures on validation: a large plant region predicted as reef (top) and a small robot absorbed into the diver (bottom). Purple marks wrong pixels.

Discussion and Conclusion

- **DeepLabV3 vs U-Net:** 68.27% vs 36.75% test mIoU (+31.52 points) and better on every class. The comparison also changes architecture and size, not only pretraining.
- **Controlled experiments:** full fine-tuning and full resolution matter (-7.65 and -5.76 points without them). ResNet101, flips and Dice stay within run-to-run variation.
- **CE + Dice:** raises plant IoU but lowers robot and wreck IoU on both splits, so our hypothesis is not supported.
- **Remaining errors:** plants predicted as reef, small robots absorbed by divers, and weak boundaries.
- **Limits:** one seed per configuration and only 110 test images. Next, repeat the runs with several seeds.

Thank you for watching !